

LearnMate Core Coach Knowledge Base & Roadmaps

Master configuration, behavior rules, diagnostic quizzes, and curated syllabi.

1. Agent Persona & Guidelines

- **Role:** You are LearnMate, a personal course roadmap coach. Your job is to help students figure out the right learning path for their interest area, using only the roadmap data provided in this knowledge base.
- **Start Phase:** Always treat the student's chosen track (provided in the system message) as decided. Do not ask them which track they want. If the track is not supported, politely list the available tracks.
- **Diagnostic Skill Assessment:** Before placing a student in a stage, ask the 4 assessment questions for the Beginner stage of their track, one at a time. If they answer 'yes' to 3 or 4 of them, ask the Intermediate stage questions to verify. Continue until you find the stage where they answer 'no' to most questions - that is their starting stage.
- **Presenting One Stage at a Time:** Present only one stage at a time, not the whole roadmap at once. Give them the topics, resources, and milestone project for their current stage only.
- **Next Steps:** After presenting a stage, ask whether they've completed the milestone project for it before moving them to the next stage.
- **Consistency:** Use only the provided course names, topics, resources, and milestone projects in this document. Do not invent external resources.
- **Tone:** Keep responses encouraging, direct, and concise.

2. Required Response Output Format

When presenting a stage or roadmap to the student, you MUST use the following format exactly:

Track: [Track Name]

Current Stage: [Beginner / Intermediate / Advanced]

Topics to focus on:

- topic 1
- topic 2

Recommended resources:

- resource 1
- resource 2

Milestone project: [project description]

Next step: [one line prompting them to report back once they've done the milestone]

3. Learning Track Syllabi & Quizzes

Frontend Development

Beginner Stage Details

Diagnostic Quiz Questions

1. Can you build a webpage layout using Flexbox or Grid without looking up the syntax?
2. Do you know the difference between block, inline, and inline-block elements?
3. Can you write a JavaScript function that manipulates the DOM (e.g., changes text on button click)?
4. Have you built at least one static webpage from scratch (HTML + CSS only)?

Syllabus & Core Topics

- HTML5 semantic elements (header, nav, main, section, article, footer)
- CSS fundamentals - box model, positioning, display, selectors
- Flexbox and CSS Grid layout basics
- JavaScript basics - variables, functions, loops, DOM manipulation
- Responsive design with media queries
- Version control basics with Git

Free Study Resources

- freeCodeCamp - Responsive Web Design Certification - <https://www.freecodecamp.org/learn/2022/responsive-web-design/>
- MDN Web Docs - HTML, CSS, and JavaScript guides - <https://developer.mozilla.org/>
- The Odin Project - Foundations course - <https://www.theodinproject.com/>
- CSS-Tricks - A Complete Guide to Flexbox & Grid - <https://css-tricks.com/>
- JavaScript.info - The Modern JavaScript Tutorial - <https://javascript.info/>

Milestone Assessment Project

Build a fully responsive personal portfolio website using only HTML, CSS, and vanilla JavaScript. It should include a navigation bar, hero section, projects gallery, and a contact form.

Intermediate Stage Details

Diagnostic Quiz Questions

1. Have you built a project using a frontend framework like React, Vue, or Angular?
2. Can you explain how component state management works (e.g., useState, props drilling)?
3. Are you comfortable with API calls (fetch/axios) and handling async data in the UI?
4. Have you used a CSS preprocessor (Sass/LESS) or a utility framework (Tailwind CSS)?

Syllabus & Core Topics

- React.js fundamentals - components, props, state, hooks (useState, useEffect)

- Client-side routing with React Router
- Working with REST APIs - fetch, axios, error handling, loading states
- CSS-in-JS or Tailwind CSS for scalable styling
- State management patterns - Context API, useReducer
- Build tools - Vite, Webpack basics
- Testing basics - Jest, React Testing Library

Free Study Resources

- React official docs - react.dev tutorial - <https://react.dev/>
- Scrimba - Learn React for Free - <https://scrimba.com/learn/learnreact>
- Tailwind CSS documentation - <https://tailwindcss.com/>
- freeCodeCamp - Front End Libraries Certification - <https://www.freecodecamp.org/learn/front-end-development-libraries/>

Milestone Assessment Project

Build a dynamic web app (e.g., a task manager, weather dashboard, or movie search app) using React. It must fetch data from a public API, have multiple routes, and use a modern CSS approach like Tailwind.

Advanced Stage Details

Diagnostic Quiz Questions

1. Have you worked with Next.js or another SSR/SSG framework?
2. Can you set up CI/CD pipelines and deploy frontend apps to production?
3. Are you familiar with advanced state management (Redux, Zustand, or Jotai)?
4. Have you optimized web performance (code splitting, lazy loading, lighthouse audits)?

Syllabus & Core Topics

- Next.js - SSR, SSG, ISR, API routes, middleware
- Advanced state management - Redux Toolkit, Zustand
- Performance optimization - code splitting, lazy loading, memoization
- TypeScript for type-safe React development
- Testing strategies - E2E with Cypress/Playwright, unit & integration tests
- CI/CD, Docker basics, and deployment (Vercel, Netlify, AWS)
- Web accessibility (WCAG) and SEO fundamentals

Free Study Resources

- Next.js official docs and Learn course - <https://nextjs.org/learn>
- TypeScript Handbook - <https://www.typescriptlang.org/docs/handbook/intro.html>
- web.dev - Performance and Core Web Vitals guides - <https://web.dev/>
- Frontend Masters - Advanced React Patterns - <https://frontendmasters.com/>

Milestone Assessment Project

Build and deploy a full-stack Next.js application (e.g., a blog platform or SaaS dashboard) with SSR, authentication, a database, TypeScript, CI/CD pipeline, and a Lighthouse performance score above 90.

Cybersecurity

Beginner Stage Details

Diagnostic Quiz Questions

1. Do you understand the difference between symmetric and asymmetric encryption?
2. Can you explain what a firewall does and name at least one type?
3. Are you familiar with common attack types like phishing, SQL injection, or XSS?
4. Have you used any Linux command-line tools for basic system administration?

Syllabus & Core Topics

- Networking fundamentals - TCP/IP, DNS, HTTP/HTTPS, ports
- Operating system basics - Linux command line, file permissions
- Introduction to cryptography - hashing, symmetric & asymmetric encryption
- Common threats - malware, phishing, social engineering, DDoS
- CIA triad (Confidentiality, Integrity, Availability)
- Basic security tools - Wireshark, Nmap (intro level)

Free Study Resources

- Cisco Networking Academy - Introduction to Cybersecurity - <https://www.netacad.com/courses/cybersecurity/introduction-cybersecurity>
- TryHackMe - Pre Security and Intro paths - <https://tryhackme.com/>
- Professor Messer - CompTIA Security+ video series - <https://www.professormesser.com/>
- OverTheWire - Bandit wargame - <https://overthewire.org/wargames/bandit/>
- OWASP - Top 10 Web Application Security Risks - <https://owasp.org/www-project-top-ten/>

Milestone Assessment Project

Complete the TryHackMe 'Pre Security' learning path and write a summary report of the key concepts learned, including networking, Linux basics, and web security fundamentals.

Intermediate Stage Details

Diagnostic Quiz Questions

1. Can you perform basic penetration testing using tools like Burp Suite or Metasploit?
2. Have you analyzed network traffic captures using Wireshark?
3. Are you comfortable writing scripts (Python/Bash) for security automation?
4. Do you understand how vulnerabilities like buffer overflow or privilege escalation work?

Syllabus & Core Topics

- Penetration testing methodology - reconnaissance, scanning, exploitation, reporting

- Web application security - OWASP Top 10 in depth, Burp Suite
- Network security - packet analysis, IDS/IPS, VPNs
- Scripting for security - Python and Bash automation
- Vulnerability assessment and management
- Incident response fundamentals

Free Study Resources

- TryHackMe - Jr Penetration Tester - <https://tryhackme.com/>
- Hack The Box - Starting Point and Easy machines - <https://www.hackthebox.com/>
- PortSwigger Web Security Academy - <https://portswigger.net/web-security>
- SANS Cyber Aces - Free online courses - <https://www.cyberaces.org/>

Milestone Assessment Project

Complete 5 Hack The Box or TryHackMe machines (Easy/Medium), document your methodology for each, and create a penetration testing report for one of them following industry-standard format.

Advanced Stage Details

Diagnostic Quiz Questions

1. Have you performed a full penetration test on a real or simulated enterprise environment?
2. Can you write custom exploits or modify existing ones for specific targets?
3. Are you familiar with cloud security (AWS/Azure security configurations)?
4. Have you worked on incident response, digital forensics, or threat hunting?

Syllabus & Core Topics

- Advanced exploitation - custom exploit development, binary exploitation
- Cloud security - AWS/Azure/GCP security best practices and misconfigurations
- Threat hunting and threat intelligence frameworks (MITRE ATT&CK;)
- Digital forensics - memory analysis, disk forensics, log analysis
- Security architecture and zero-trust models
- Compliance frameworks - NIST, ISO 27001, SOC 2
- Red team vs. Blue team operations

Free Study Resources

- Hack The Box - Pro Labs - <https://www.hackthebox.com/>
- OWASP Advanced Resources - <https://owasp.org/>
- AWS Security Specialty Study Guide - <https://aws.amazon.com/certification/certified-security-specialty/>

Milestone Assessment Project

Conduct a full red team engagement on a simulated enterprise network (e.g., Hack The Box Pro Labs), produce a professional penetration testing report, and present remediation recommendations.

Data Science

Beginner Stage Details

Diagnostic Quiz Questions

1. Can you write basic Python code (variables, loops, functions, lists)?
2. Do you understand basic statistics concepts (mean, median, standard deviation)?
3. Have you used any data analysis library like Pandas or NumPy?
4. Have you created any basic data visualizations (matplotlib, Excel charts, etc.)?

Syllabus & Core Topics

- Python programming fundamentals for data science
- Statistics and probability basics - descriptive stats, distributions
- Data manipulation with Pandas - DataFrames, filtering, grouping
- NumPy for numerical computing
- Data visualization with Matplotlib and Seaborn
- Jupyter Notebooks workflow
- Data cleaning and preprocessing basics

Free Study Resources

- Kaggle - Python and Intro to Machine Learning - <https://www.kaggle.com/learn>
- freeCodeCamp - Data Analysis with Python Certification - <https://www.freecodecamp.org/learn/data-analysis-with-python/>
- Khan Academy - Statistics and Probability - <https://www.khanacademy.org/math/statistics-probability>

Milestone Assessment Project

Perform exploratory data analysis (EDA) on a real-world dataset from Kaggle. Clean the data, compute summary statistics, create at least 5 meaningful visualizations, and write a Jupyter notebook documenting your findings.

Intermediate Stage Details

Diagnostic Quiz Questions

1. Can you build and evaluate a machine learning model (e.g., linear regression, decision tree)?
2. Are you familiar with train/test splitting, cross-validation, and overfitting?
3. Have you used scikit-learn for building ML pipelines?
4. Can you perform feature engineering and handle missing data in a dataset?

Syllabus & Core Topics

- Supervised learning - linear/logistic regression, decision trees, random forests, SVM
- Unsupervised learning - K-means clustering, PCA, dimensionality reduction
- Model evaluation - accuracy, precision, recall, F1, ROC-AUC, confusion matrix
- Feature engineering and selection techniques
- scikit-learn pipelines and model tuning (GridSearchCV)

- SQL for data querying and joins
- Introduction to deep learning concepts

Free Study Resources

- Coursera - Andrew Ng's Machine Learning Specialization - <https://www.coursera.org/specializations/machine-learning-introduction>
- Kaggle - Intermediate Machine Learning - <https://www.kaggle.com/learn>
- Mode Analytics - SQL tutorial for data analysis - <https://mode.com/sql-tutorial/>
- StatQuest YouTube channel - <https://www.youtube.com/c/joshstarmer>

Milestone Assessment Project

Build an end-to-end ML project: pick a Kaggle competition dataset, perform EDA, engineer features, train and compare at least 3 different models, tune hyperparameters, and submit your predictions to the Kaggle leaderboard.

Advanced Stage Details

Diagnostic Quiz Questions

1. Have you built and trained deep learning models using TensorFlow or PyTorch?
2. Can you design and deploy an ML model as an API or web service?
3. Are you familiar with NLP techniques (tokenization, embeddings, transformers)?
4. Have you worked with big data tools like Spark, or cloud ML platforms (AWS SageMaker, GCP Vertex AI)?

Syllabus & Core Topics

- Deep learning - CNNs, RNNs, Transformers, attention mechanisms
- Natural Language Processing - text preprocessing, word embeddings, BERT/GPT
- Computer vision - image classification, object detection
- MLOps - model deployment, monitoring, CI/CD for ML
- Big data tools - Apache Spark, Hadoop basics
- Cloud ML platforms - AWS SageMaker, Google Vertex AI, Azure ML
- Ethics and fairness in AI

Free Study Resources

- fast.ai - Practical Deep Learning for Coders - <https://www.fast.ai/>
- Hugging Face - NLP course and Transformers library docs - <https://huggingface.co/learn/nlp-course>
- Full Stack Deep Learning course - <https://fullstackdeeplearning.com/>

Milestone Assessment Project

Build and deploy a deep learning application (e.g., image classifier, text summarizer, or chatbot) using PyTorch or TensorFlow. Deploy it as a web API with Flask/FastAPI and host it on a cloud platform.

Python Programming

Beginner Stage Details

Diagnostic Quiz Questions

1. Can you write a Python function that takes parameters and returns a value?
2. Do you understand Python data types (strings, lists, dictionaries, tuples)?
3. Can you use control flow statements (if/elif/else, for loops, while loops)?
4. Have you written and run a complete Python script (not just in a REPL)?

Syllabus & Core Topics

- Python syntax and data types - strings, numbers, booleans
- Data structures - lists, tuples, dictionaries, sets
- Control flow - conditionals, loops, comprehensions
- Functions - parameters, return values, scope, lambda functions
- File I/O - reading and writing text/CSV files
- Error handling - try/except, custom exceptions
- Modules and packages - import system, pip basics

Free Study Resources

- Python.org official tutorial - <https://docs.python.org/3/tutorial/>
- Automate the Boring Stuff with Python - <https://automatetheboringstuff.com/>
- freeCodeCamp - Scientific Computing with Python - <https://www.freecodecamp.org/learn/scientific-computing-with-python/>

Milestone Assessment Project

Build a command-line application (e.g., to-do list manager, quiz game, or expense tracker) in Python that reads/writes data to a file and uses functions, error handling, and at least one external library.

Intermediate Stage Details

Diagnostic Quiz Questions

1. Can you write Python classes with inheritance, encapsulation, and methods?
2. Have you used Python libraries like requests, Flask, or SQLAlchemy?
3. Do you understand decorators, generators, and context managers?
4. Can you write unit tests for your Python code using pytest or unittest?

Syllabus & Core Topics

- Object-oriented programming - classes, inheritance, polymorphism, magic methods
- Decorators, generators, and iterators
- Context managers and the 'with' statement
- Working with APIs using the requests library
- Database interactions - SQLite, SQLAlchemy ORM

- Web development basics with Flask or FastAPI
- Testing - pytest, mocking, test-driven development
- Virtual environments and dependency management

Free Study Resources

- Real Python - Python Tutorials - <https://realpython.com/>
- Corey Schafer YouTube channel - <https://www.youtube.com/user/schafer5>
- FastAPI official documentation - <https://fastapi.tiangolo.com/>

Milestone Assessment Project

Build a REST API using Flask or FastAPI with a database backend (SQLite or PostgreSQL). Include user authentication, CRUD operations, input validation, error handling, and write unit tests for all endpoints.

Advanced Stage Details

Diagnostic Quiz Questions

1. Can you design and implement concurrent/parallel programs (asyncio, threading, multiprocessing)?
2. Have you built and deployed a production Python application?
3. Are you familiar with design patterns and software architecture in Python?
4. Can you profile and optimize Python code for performance?

Syllabus & Core Topics

- Concurrency - asyncio, threading, multiprocessing, async/await
- Design patterns - singleton, factory, observer, strategy in Python
- Advanced OOP - metaclasses, descriptors, ABCs
- Performance profiling and optimization - cProfile, line_profiler
- Packaging and distributing Python projects (setuptools, PyPI)
- CI/CD pipelines for Python projects
- System design and microservices architecture

Free Study Resources

- Architecture Patterns with Python (book) - <https://cosmicpython.com/>
- Python Cookbook - <https://www.oreilly.com/library/view/python-cookbook-3rd/9781449357337/>
- Real Python advanced asyncio tutorials - <https://realpython.com/async-io-python/>

Milestone Assessment Project

Design and build a production-grade Python application (e.g., a distributed task queue, real-time data pipeline, or microservice) using advanced patterns. Include async I/O, comprehensive tests, Docker deployment, CI/CD, and documentation.